TRADERPRO ALGORITHMIC ENGINE

ICT Strategies & Architecture Complete Working Manual

Full Technical Specification, Algorithmic Execution Rules, Multi-Timeframe Confluence, and Goat Funded Trader ($1,000 Model) Compliance Guide

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Target Prop Firm:	Goat Funded Trader ($1,000 Challenge Model)	**Primary Assets:**	NAS100 (US100), SPX500, GOLD (XAUUSD)
Risk Per Trade:	$2.50 (Strict 0.25% Balance Risk)	**Minimum R:R:**	1:2.00 (Standardized across all strategies)
Consistency Rule:	Max 15% Single-Trade Profit Cap	**Data Feed:**	TradingView WebSocket (1m/5m/15m/30m/1h/4h/1D/1W)
SMT Pairings:	NAS100 vs SPX500 | NQ vs MES (No Gold)	**Engine Status:**	**LIVE PRODUCTION (Online & Verified)**

EXECUTIVE SUMMARY & MANDATE

This document provides the definitive, comprehensive algorithmic manual for all trading strategies deployed inside the TraderPro algorithmic ecosystem. Following the September 2026 repo-wide refactor, all strategies have been decoupled from synthetic brokers, legacy scrapers, and hardcoded values. Every strategy is an authentic mathematical function of multi-timeframe TradingView data, synchronized SMT divergence, centralized ATR stop-loss buffers, and strict prop-firm capital protection. This manual details the exact internal working, setup conditions, checklist verifications, entry/exit mechanics, and invalidation rules for each strategy.

Section	Title & Core Focus	Target Markets	Execution Timeframe
Section 1	Unified Core Architecture, Risk Engine & Data Pipeline	All Instruments	Infrastructure
Section 2	Strategy 1: SMC Liquidity Master (ICT + Goat $1k)	NAS100 / GOLD	15M Setup -> 1M Execution
Section 3	Strategy 2: ICT Market Maker Model (24-Lesson Mastery)	NAS100 / SPX500	HTF -> 1M Execution
Section 4	Strategy 3: Praveen's ICT Strategy v2.1 (5-Step Flow)	NAS100 / NQ / SPX	30M Zone -> 1M CISD/IFVG
Section 5	Strategy 4: SPEAN Strategy (Q-Zone + 7-Layer FAH)	NAS100 / GOLD	15M Sweep -> 1M FAH POI
Section 6	Strategy 5: ICT Silver Bullet (Time-Based FVG)	NAS100 / GOLD	60-Min Killzone (1M)
Section 7	Strategy 6: Session Momentum Breakout	NAS100 / GOLD	15M Range Expansion
Section 8	Centralized Order Ticket & Goat Consistency Safeguards	All Instruments	Execution Contract

SECTION 1: UNIFIED ARCHITECTURE & INFRASTRUCTURE

1.1 Single Live Data Pipeline (TradingView WebSocket)

All strategies operate strictly as pure deterministic functions of a standardized market_data dictionary supplied by backend/market_data.py. Under zero circumstances does any strategy fetch data directly or maintain private HTTP scrapers. The architecture eliminates Yahoo Finance entirely in favor of an institutional WebSocket feed.

- **Direct Connection:** Subscribes to wss://data.tradingview.com/socket.io/websocket with keep-alive heartbeats and auto reconnection.
- **Exchange Tickers:** NAS100 uses CAPITALCOM:US100; SPX500 uses CAPITALCOM:SPX500; GOLD uses CAPITALCOM:GOLD.
- **8 Native Resolutions:** Subscribes concurrently to 1m, 5m, 15m, 30m, 60m (1h), 240m (4h), 1D (Daily), 1W (Weekly). Zero candle emulation.
- **Thread Safety:** Protected by threading.Lock() to ensure race-free evaluation across background tasks.
- **Timestamp Alignment:** Through align_series(), correlated pairs are indexed by exact UTC minute timestamps before SMT analysis.

1.2 Centralized Risk Engine (Goat $1,000 Challenge Model)

Prop firm survival demands mathematically rigorous capital management. All six strategies import risk calculations, position sizing, and order tickets directly from backend/risk_engine.py.

Parameter	Value / Standard	Rationale & Rule Governance
Account Balance	$1,000.00 Base	Goat Funded Trader micro-tier account size.
Risk Per Trade	$2.50 (0.25%)	Conservative sizing allows 12 consecutive losses before hitting 3% daily DD limit.
Minimum R:R	1:2.00 (Strict)	Any setup yielding < 2.0 R:R is automatically discarded ('Wait / No trade').
Consistency Profit Cap	15% Rule	No single trade may constitute > 15% of the total profit target ($150 target = $22.50 cap).
Consistency Formula	(Profit / Balance) * 100	Genuine balance calculation (replaces legacy no-op (p/100)*100).
SL Buffer: NAS100	3.0 Points	Protects against broker spread fluctuation and institutional wick slippage.
SL Buffer: SPX500	1.5 Points	Calibrated to S&P; 500 tick size and typical 1M spread.
SL Buffer: GOLD	1.0 Point ($0.10)	Accounts for spot gold volatility and broker spread spikes.

1.3 Smart Money Technique (SMT) Correlated Engine

SMT Divergence detects when two correlated assets in the same class diverge at key swing points, revealing genuine institutional accumulation or distribution. The centralized backend/smt_engine.py enforces strict rules:

- **Permitted Correlated Pairs:** NAS100 <-> SPX500 and NQ <-> MES (US equity indices sharing macroeconomic drivers).
- **Strict Commodity Exclusion:** GOLD is strictly prohibited from equity index SMT comparisons (eliminating false signals).
- **Bullish SMT Divergence:** Primary asset makes a Lower Low (sweeps liquidity) while correlated asset holds a Higher Low (institutional accumulation).
- **Bearish SMT Divergence:** Primary asset makes a Higher High (sweeps liquidity) while correlated asset holds a Lower High (institutional distribution).

SECTION 2: STRATEGY 1 — SMC LIQUIDITY MASTER (ICT + GOAT $1K)

2.1 Core Theoretical Concept

SMC Liquidity Master is a multi-timeframe institutional model that capitalizes on stop runs outside established 15-minute ranges. Retail traders place stops above swing highs (Buy-Side Liquidity - BSL) and below swing lows (Sell-Side Liquidity - SSL). Smart money sweeps this liquidity to fill large institutional orders, followed by immediate displacement in the opposite direction.

SMC LIQUIDITY MASTER EXECUTION FLOW

`[15M Range Filter]` → `[BSL/SSL Sweep]` → `[1M BOS Displacement]` → `[50% POI Retracement]` → `[1:2+ R:R Target]`

2.2 Multi-Timeframe Alignment Sequence

Step 1: 4H Structural Bias Filter: Analyzes the last 12 four-hour candles (48 hours). If price is in the upper half of the 4H range, market is in Premium; if in the lower half, Discount. Longs are favored in Discount; Shorts in Premium.

Step 2: 1H Session Context & PDH/PDL Distinction: Prior Day High (PDH) and Low (PDL) are strictly sourced from completed prior daily bar (candles_1d[-2]). Today's intraday high and low are tracked independently as *Today's Intraday Range*, permanently resolving the PDH/PDL overwrite bug.

Step 3: 15-Minute Liquidity Sweep: Identifies unmitigated 15M swing highs (BSL) and lows (SSL) over a 20-candle lookback (5 hours). A sweep occurs when price pierces the 15M level by at least 0.2 points, absorbing liquidity.

Step 4: 1-Minute Break of Structure (BOS): Price must demonstrate aggressive displacement back inside the range, breaking the most recent 1M swing low (Bearish BOS) or 1M swing high (Bullish BOS). The sweep extreme becomes the **Protected Level**.

Step 5: Point of Interest (POI) & Limit Entry: A 1M Fair Value Gap or Supply/Demand zone is formed between the Protected Level and displacement close. A Limit Order is placed at the 50% midpoint of this zone.

Component	Bullish Setup (Buy Limit)	Bearish Setup (Sell Limit)
Pre-Condition	15M Sell-Side Liquidity (SSL) swept	15M Buy-Side Liquidity (BSL) swept
1M Confirmation	Price breaks above recent 1M swing high (Bullish BOS)	Price breaks below recent 1M swing low (Bearish BOS)
Protected Level	Protected LOW = Lowest wick of the SSL sweep	Protected HIGH = Highest wick of the BSL sweep
Entry POI	50% midpoint between Protected Low & current price	50% midpoint between Protected High & current price
Stop Loss	**Protected Low - sl_buffer(symbol)**	**Protected High + sl_buffer(symbol)**
Take Profit	**Target opposing 15M BSL (capped at 1:3.0 R:R)**	**Target opposing 15M SSL (capped at 1:3.0 R:R)**
Session Filter	12:00 – 20:00 UTC (Pre-NY and NY Active)	12:00 – 20:00 UTC (Pre-NY and NY Active)
Cancel Invalidation	If price breaches Protected Low or hits TP before fill	If price breaches Protected High or hits TP before fill

SECTION 3: STRATEGY 2 — ICT MARKET MAKER MODEL (24-LESSON MASTERY)

3.1 Core Theoretical Concept

The ICT Market Maker Model (MMBM / MMSM) synthesizes Michael J. Huddleston's complete 24-lesson core curriculum into an autonomous decision engine. The model reflects how institutional algorithmic market makers inject liquidity, manipulate retail sentiment during specific killzones, engineer liquidity legs, and distribute inventory into HTF PD Arrays.

ICT MARKET MAKER AMD TIMING CYCLE

```
[Asian Accumulation (20-00 NY)] → [London Judas Sweep (02-05 NY)] → [NY Distribution (07-10 NY)] →
                                                                    [HTF PD Array Target]
```

3.2 Key Algorithmic Pillars (Lessons 1 – 24)

1. **Weekly Bias Filter (Lesson 8):** Prior completed weekly bar range (candles_1w[-2]). If trading above prior week midpoint (Equilibrium), Weekly Bias is Bullish; if below, Bearish.
2. **Time & Price Midnight Open (Lesson 9):** Tracks 00:00 NY Midnight opening price. Buying is favored strictly at Discount below Midnight Open; Selling is favored at Premium above Midnight Open.
3. **AMD / Power of Three (Lesson 10):** Cycles through Accumulation (Asian consolidation), Manipulation (Judas swing stop sweep), and Distribution (multi-candle displacement towards target).
4. **Asian Range Measurement & Judas Swing (Lessons 11 & 12):** Asian range established between 20:00 and 00:00 NY. London false break sweeping Asian High followed by Bullish MSS triggers a verified Judas Reversal.
5. **HTF PD Array Confluence (Lessons 6 & 23):** Price must reach or tap an authentic 4H, 1H, or 30M unmitigated Order Block or Fair Value Gap before LTF execution is permitted.
6. **Displacement & MSS (Lessons 3 & 14):** Market Structure Shift requires 1M candle bodies to exceed the swing point with body size > 1.4x the 20-period average candle body.
7. **SMT Divergence Booster (Lesson 16):** Confirmed SMT divergence between NAS100 and SPX500 acts as an institutional confluence booster, raising setup confidence.
8. **Optimal Trade Entry (OTE) (Lesson 18):** Identifies Fibonacci retracement sweet spot: 61.8% to 79.0% (optimal: 70.5%) of the displacement swing.

3.3 Master Checklist Matrix (11 Confluence Checks)

#	ChecklistItem	Verification Condition	Threshold / Source
1	HTF Bias & Trend	4H/1H swing alignment + Weekly equilibrium check	c_4h, c_1h, c_1w
2	Time & Price (Daily Open)	Discount below 00:00 NY open for Buy; Premium for Sell	NY 00:00 EST Open
3	Session & Killzone	Inside Asian (20-00), London (02-05), NY (07-10), Close (10-12)	America/New_York
4	AMD Phase	State transition: Accumulation -> Manipulation -> Distribution	ATR compression + KZ
5	Judas Swing Detection	Asian extreme swept during London/NY followed by MSS	Asian High/Low
6	HTF PD Array	Price tapping 4H/1H/30M Order Block or Fair Value Gap	Unmitigated OB/FVG
7	Liquidity Hunt (Sweep)	External liquidity pool swept before reversal	BSL / SSL Pool
8	Market Structure Shift	Displacement close breaking key swing point (> 1.4x body)	1M Candle Series
9	Entry FVG & OTE Zone	Unfilled 1M FVG formed by displacement leg	61.8% - 79.0% Fib
10	SMT Divergence	NAS100 vs SPX500 non-confirmation at swing extremes	smt_engine
11	Execution & Risk ($1k)	$2.50 Risk (0.25%), Min 1:2.0 R:R, < 15% Consistency Cap	risk_engine

3.4 Dual-Target Exit Framework (Lessons 15 & 19)

The Market Maker strategy implements a two-stage profit exit: **Take Profit 1 (Scale)** is set at 1.5x risk to secure partial capital and move stops to breakeven; **Take Profit 2 (Final DOL)** targets opposing major liquidity pools or 2.5x expansion, maintaining Goat prop firm compliance.

SECTION 4: STRATEGY 3 — PRAVEEN'S ICT STRATEGY V2.1 (5-STEP FLOW)

4.1 Overview & Operational Rules

Praveen's ICT Strategy v2.1 is a high-precision scalping framework designed for NAS100/NQ using 30M PD Arrays for location and 1M execution confirmation via Change in State of Delivery (CISD) and Inversion Fair Value Gaps (IFVG). Trading is restricted to a maximum of 3 trades per day (1 per session: London, NY AM, NY PM).

PRAVEEN 5-STEP SCALPING ENGINE

`[1. HTF Bias & 00:00 Open] → [2. DOL Target] → [3. 30M PD Array Tap] → [4. 1M CISD + IFVG] → [5. 50% CE Limit Entry]`

4.2 The 5-Step Sequential Entry Engine

Step 1: BIAS Determination: Evaluates HTF flow across Weekly -> Daily -> 4H -> 1H. Bullish bias requires trading above Weekly Equilibrium, Daily Bias Bullish, and price below Midnight Open (Discount). Bearish bias requires trading below Weekly Equilibrium, Daily Bias Bearish, and price above Midnight Open (Premium).

Step 2: Draw on Liquidity (DOL) Mapping: Smart money moves toward resting liquidity. The engine ranks DOL targets: 1. Equal Highs/Lows (EQH/EQL); 2. Prior Day High/Low (PDH/PDL); 3. Prior Week High/Low (PWH/PWL); 4. Major 1H/4H Swings. Trade MUST have un-swept DOL in direction of bias.

Step 3: Internal Range Liquidity (IRL): Entry zone must be an authentic 30-Minute Fair Value Gap (FVG) or Order Block (OB). Price must tap into the 30M zone before 1M setup is monitored. Anticipating or front-running is prohibited.

Step 4: 1-Minute Entry Flow (LRLR + CISD + IFVG):
- *LRLR Path Check:* Verifies a clean path of Low Resistance Liquidity Run with no High Resistance blockers.
- *CISD (Change in State of Delivery):* The first candle closing beyond open of displacement candle pushing into zone.
- *IFVG (Inversion FVG):* A prior opposing 1M FVG broken with displacement, inverting into support/resistance.

Step 5: SL & TP Placement (Consequent Encroachment):
- *Entry:* Limit order at Consequent Encroachment (C.E. 50% midpoint) of the 1M IFVG.
- *Stop Loss:* Beyond the swing extreme that created the CISD + sl_buffer(symbol) (3.0 pts for NAS100).
- *Take Profit:* Drawn directly to Step 2 DOL target, strictly enforcing a minimum of 1:2.00 R:R.

4.3 Quarterly Theory & SSMT (90-Minute Cycles)

Divides daily sessions into 90-minute quarters (Q1 Accumulation, Q2 Manipulation, Q3 Distribution, Q4 Reversal/Continuation). **SSMT Divergence** verifies SMT non-confirmation on 5M/15M bars at quarter cycle boundaries between NAS100 and SPX500.

4.4 Setup Grading Matrix

Grade	Action	Criteria Met	Execution Mandate
A+ Setup	Execute Now	All 5 Steps + SMT Divergence + LRLR Clear Path	Full Goat Risk ($2.50). High conviction.
A Setup	Execute	All 5 Steps met; SMT aligned in direction	Full Goat Risk ($2.50). Standard execution.
B Setup	Proceed with Caution	Minor HRLR blocker or secondary 15M zone	Reduce risk or wait for second CISD confirmation.
C Setup	Do Not Trade	Counter-bias, missing 30M tap, or R:R < 2.0	Stand aside. Wait / No trade.

SECTION 5: STRATEGY 4 — SPEAN STRATEGY (7-LAYER FAH CONFLUENCE)

5.1 Overview & Philosophy

SPEAN Strategy is an ultra-disciplined price action model based on the concept of the **FAH (First Candle That Closed Beyond Swing Extreme)**. It eliminates subjective interpretation by requiring a strict 7-layer sequential confluence filter before any order is submitted. If even a single layer fails, the algorithm stands aside.

SPEAN 7-LAYER GATEKEEPER SEQUENCE

```
[1. Q-Zone Window] → [2. 15M Sweep] → [3. FAH Extreme] → [4. Displacement] → [5. 1M POI] → [6. Retracement] → [7. 1:2+ R:R]
```

5.2 The 7-Layer Sequential Execution Framework

Layer 1: Time & Session Window (Q-Zone): Restricted to London Q-Zone (07:00 – 10:00 UTC) and NY Q-Zone (12:00 – 16:00 UTC). Trades barred during lunch and Asian quiet hours.

Layer 2: 15-Minute Liquidity Take: Market must sweep a significant 15M structural extreme: Asian High/Low, Prior Day High/Low (PDH/PDL), or Today's Intraday Extreme.

Layer 3: FAH Confirmation (First Candle Closed Beyond): Core signature of SPEAN. Identifies the specific 1M candle that achieved the first confirmed close beyond the swept level. Its high and low define the active range.

Layer 4: 1-Minute Strong Displacement: Price aggressively rejects the swept level, producing a strong displacement candle closing completely back inside the FAH range with expanded volume.

Layer 5: Valid Point of Interest (POI): Displacement must leave an unmitigated 1M Fair Value Gap or Order Block inside the displacement leg, establishing the institutional footprint.

Layer 6: Retracement to POI Entry: Price must retrace back into the POI zone. A Limit Order is placed at the proximal boundary of the POI. No market chasing is permitted.

Layer 7: Goat Risk & 1:2+ R:R Validation: Stop Loss placed beyond FAH extreme + sl_buffer(symbol). Take Profit targets opposing session liquidity with minimum 1:2.00 R:R. Consistency verified under 15% cap.

SPEAN INVALIDATION CRITERIA

SPEAN Invalidation Rule: An active setup is invalidated immediately if price closes beyond the FAH Extreme before reaching the entry, or if price reaches the Take Profit target without filling the limit order. This ensures the algorithm never enters a stale or structurally compromised setup.

SECTION 6: STRATEGY 5 — ICT SILVER BULLET (TIME-BASED FVG)

6.1 Core Theoretical Concept

The ICT Silver Bullet operates on the mathematical certainty that algorithmic market makers deliver high-probability repricing during three specific 60-minute daily windows. Within these exact 60 minutes, the algorithm hunts for liquidity sweeps followed by Fair Value Gap expansion.

Silver Bullet Window	New York Local Time	UTC Time	Market Focus & Volatility
London Silver Bullet	03:00 AM – 04:00 AM NY	08:00 – 09:00 UTC	London morning expansion post-Asian sweep.
New York AM Silver Bullet	10:00 AM – 11:00 AM NY	14:00 – 15:00 UTC	Post-equity open liquidity run (highest volume).
New York PM Silver Bullet	02:00 PM – 03:00 PM NY	18:00 – 19:00 UTC	Afternoon bond close / institutional rebalance.

6.2 Step-by-Step Execution Sequence

Step 1: Liquidity Take: Market sweeps an external liquidity pool (BSL, SSL, PDH, PDL, or session extreme) prior to or inside the window.

Step 2: 1M Displacement & FVG: A displacement candle inside the 60-minute window forms a clean 3-candle FVG (>= 3.0 pts for NAS100; >= 0.8 pts for GOLD).

Step 3: Limit Order Placement: Limit order placed at proximal boundary of the FVG (top of gap for Buy; bottom of gap for Sell).

Step 4: Stop Loss Placement: Placed at displacement swing low (Buy) or high (Sell) with sl_buffer(symbol) applied.

Step 5: Target DOL: Target placed at opposing 15M/1H swing pool, guaranteeing a minimum of 1:2.00 R:R.

Step 6: Window Invalidation: If the 60-minute window closes without a fill, the pending limit order is automatically canceled.

SECTION 7: STRATEGY 6 — SESSION MOMENTUM BREAKOUT

Session Momentum Breakout monitors the established 4-hour session range (last 16 candles on 15-minute timeframe). When market volume accelerates and cleanly displaces beyond the range boundary with a volatility buffer (5.0 pts for NAS100; 0.8 pts for GOLD), a Stop/Market order is triggered targeting a 1:2.00 momentum expansion.

Stop Loss & Invalidation: Stop loss is positioned just inside the broken range boundary using sl_buffer(symbol). If price retreats back inside the session range, the trade is canceled immediately to protect Goat capital.

SECTION 8: STANDARDIZED ORDER TICKET & SAFEGUARDS

8.1 Master Order Ticket Specification

Whenever any strategy confirms an active execution setup, it generates an immutable, Goat-compliant order ticket through build_order_ticket(). This ticket is broadcast simultaneously over WebSocket and Telegram.

Field Name	Sample Value (NAS100 Buy)	Operational Function
Order Type	**BUY LIMIT**	Execution method (Limit / Stop / Market)
Instrument	NAS100	Target trading asset
Entry Price	30,631.7	Exact limit entry price (CE / FVG boundary)
Stop Loss	**30,621.9**	Calculated invalidation price + buffer
Take Profit	**30,651.3**	Draw on Liquidity (DOL) target price
R:R Ratio	**1:2.00**	Strict minimum 1:2.0 enforced
Risk Amount	$2.50 (0.25%)	Goat $1,000 challenge risk per trade
Projected Profit	**+$5.00**	Anticipated dollar gain upon reaching TP
Consistency Impact	0.50% (< 15% Cap Safe)	Verified against Goat 15% single-trade rule
Cancel If	Price closes beyond 30,621.9 or sweeps TP	Automated order cancellation condition

8.2 Goat Funded Trader Consistency Safeguards

Prop firms enforce strict consistency rules to eliminate 'lucky gambles'. In a $1,000 account targeting a $150 profit, no single trade may exceed $22.50 (15% of total profits). Our risk engine computes:

TRUE CONSISTENCY PROFIT CAP FORMULA

Mathematical Formula:

Consistency % = (Projected Profit / Account Balance) × 100

*Example: $5.00 projected profit on a $1,000.00 account = **0.50%** (< 15% Cap Safe). Even on a max 1:5.0 R:R windfall ($12.50 gain), consistency impact is only 1.25%, keeping every trade safely below prop firm disqualification.*

SYSTEM COMPLIANCE VERIFICATION

VERIFICATION & DEPLOYMENT NOTE: All logic documented in this manual has been compiled, unit-tested across synthetic and live market environments, and deployed to live production on Railway. The system runs 24/7 with zero Yahoo Finance scraping and sub-second TradingView WebSocket delivery.